ARF Portal: A Web-Based Veterinary Appointment and Management Information System for KHO Veterinary Clinic

"Caring Smarter, Healing Better"

A Collaborative Project for Web Systems and Technologies and Management Information Systems Subjects

Information Technology Program STI College Naga

Submitted by: Team Vertex

Group Leader: Gabriel S. Bantigui

Member: Noel Andrei N. Delumen

Member: Mark Anthony V. Vergara

Submitted to:

IT Professor: Everild Gerd R. Pablo, MBA

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CHAPTER I

A. INTRODUCTION

The veterinary services industry, particularly small and medium-scale clinics such as KHO Veterinary Clinic, continues to experience growing demand driven by the increasing number of pet owners and the rising awareness of animal health and welfare across the Philippines. As veterinary clinics expand their client base and diversify their service offerings including check-ups, grooming, dental care, and vaccination programs the complexity of daily operations such as appointment scheduling, patient record management, and clinical reporting grows substantially. Traditional paper-based systems, still widely prevalent among small veterinary practices in the country, have been consistently shown to be inefficient, error-prone, and incapable of scaling with operational demand, resulting in delayed record retrieval, scheduling conflicts, and inaccurate clinical documentation.

These challenges are not unique to a single clinic. Research in both local and international contexts confirms that service-oriented healthcare practices operating without a centralized digital system experience fragmented patient recordkeeping, poor appointment coordination, and recurring administrative bottlenecks that directly affect both service quality and client satisfaction. The absence of real-time data visibility across clinical operations makes it exceedingly difficult for clinic owners and veterinarians to make timely and informed decisions on scheduling, resource allocation, and service management. This study proposes the ARF Portal, a web-based veterinary appointment and management information system, designed to digitalize clinical recordkeeping, automate appointment scheduling, and support data-driven decision-making for KHO Veterinary Clinic, Naga City.

Background of the Study

The KHO Veterinary Clinic currently relies on a traditional paper-based approach and simple spreadsheets for patient registration, appointment tracking, and medical consultations. While this manual approach was adequate during the clinic's early stages, it has become progressively inefficient as the client base continues to grow. The reliance on physical documentation poses serious threats to data integrity and leads to administrative congestion, particularly during peak seasons when retrieval of records becomes chaotic and time-consuming.

The absence of a centralized digital system means that patient histories, vaccination records, and service schedules are stored in disjointed paper files and informal spreadsheets. This fragmentation not only slows down the consultation process but also creates a higher risk of data loss, human error in record-keeping, and miscommunication between veterinarians and front desk staff. The clinic has reached a point where the volume of operations exceeds what manual systems can efficiently handle.

To address these operational challenges, this project introduces the ARF Portal, a web-based veterinary appointment and management information system designed to centralize the clinic's entire workflow. This digital transformation will enable real-time access to health records, automate appointment scheduling, provide financial and service analytics, and improve communication with pet owners through organized and accurate data management. The ARF Portal aims to bring KHO Veterinary Clinic's operations into the modern digital age, improving both service delivery and administrative efficiency.

Problem Statement

The following major problems were identified based on the existing manual processes at KHO Veterinary Clinic:

1. Delayed Record Retrieval: Searching for physical paper files of returning patients consumes significant time, slowing down the consultation and admission process, especially during busy periods.
2. Lack of Real-Time Appointment Monitoring: Scheduling is prone to overbooking or missed slots due to the absence of a centralized digital calendar, leading to client dissatisfaction and operational inefficiency.
3. Difficult Report Generation: Compiling monthly sales and patient statistics requires days of manual calculation, making it difficult for management to make timely, data-driven decisions.
4. Data Redundancy and Integrity Issues: Without a unified system, the same patient information may be recorded across multiple files, resulting in data duplication and inconsistencies.
5. Inefficient Service Tracking: The clinic lacks a proper mechanism to track the full history of services rendered per pet (e.g., vaccinations, grooming, dental procedures), making it harder for veterinarians to deliver accurate, personalized care.

Research Objectives

The primary objective of this study is to design and develop a web-based management information system for KHO Veterinary Clinic that digitalizes patient medical records, automates appointment scheduling, and generates real-time analytical reports to significantly improve the clinic's overall operational efficiency and service quality.

1. To implement a centralized scheduling module for managing appointments related to check-ups, grooming, dental services, and vaccinations, eliminating the risk of overbooking.
2. To develop a digital patient profile system that stores complete pet and owner information, enabling quick retrieval of medical histories and service records.
3. To generate automated monthly and yearly summary reports for revenue tracking and patient visit analytics, reducing manual computation time.
4. To provide an analytics dashboard that visually displays peak business hours, most popular services, and service trends to support proactive management decisions.
5. To reduce client waiting time and administrative bottlenecks through streamlined digital data processing across all user roles.

Scope and Limitations

The ARF Portal is designed to cover the core operational areas of KHO Veterinary Clinic, providing a comprehensive digital solution for its day-to-day clinical and administrative needs. The system supports patient management through the creation and maintenance of digital profiles for pets and their owners, encompassing contact details, breed information, and complete service histories. It also includes service tracking capabilities for digitally logging all clinical services rendered, such as vaccinations, surgeries, grooming sessions, dental procedures, and prescription records. An integrated appointment system enables online booking and calendar management across all service categories, featuring real-time availability monitoring and conflict detection to prevent scheduling errors. The system further implements user role management with defined access levels for Admins, Veterinarians, and Front Desk Staff, ensuring that each user interacts only with the modules and data relevant to their responsibilities. Additionally, the ARF Portal incorporates MIS reporting features that automate the generation of financial summaries, service frequency analytics, and dashboard visualizations to support informed management decision-making.

The ARF Portal, however, operates within several defined limitations that establish the boundaries of the current system version. The platform is developed exclusively as a web-based application and does not include a mobile counterpart for iOS or Android devices. Financial transactions are limited to cash and walk-in arrangements only, as the system does not support online payment processing of any kind. The scope of deployment is confined to a single branch of KHO Veterinary Clinic and does not accommodate multi-branch operations in its present form. Furthermore, the management of medicine and supply inventory is not covered by the current version of the system. Lastly, the ARF Portal requires a stable internet connection for full and uninterrupted functionality, as it does not support offline access or local-only operation.

Significance of the Study

The transition of KHO Veterinary Clinic from a manual, paper-based process to a centralized digital platform represents a significant advancement for all stakeholders involved. By focusing on digital record-keeping and streamlined scheduling, the ARF Portal delivers the following benefits:

1. To Clinic Management (Admin/Owner): The system provides a reliable tool for data-driven decision-making through automated analytics of service trends, such as peak grooming or vaccination periods. It reduces administrative overhead and minimizes financial inaccuracies caused by human error, allowing owners to focus on business growth rather than paperwork.
2. To Veterinarians: Immediate access to accurate and complete medical histories including past dental work, immunization records, and treatment notes enables quicker, more accurate diagnoses and more personalized care for every animal patient.
3. To Front Desk Staff: The system eliminates administrative congestion by simplifying registration, appointment booking, and patient lookup. Daily operations become more organized, reducing the likelihood of overbooking and scheduling errors.
4. To Pet Owners and Patients: Clients benefit from reduced waiting times, more accurate service records, and an overall improvement in the quality of veterinary care provided to their pets.
5. To Future Researchers: This study serves as a working model and reference point for digitizing specialized service-oriented clinical operations. It provides a practical foundation for future web-based management systems in small-scale veterinary or similar healthcare practices.

Front Desk Staff: Creates client and patient records, schedules appointments, searches records, and updates appointment statuses during daily clinic operations.
Veterinarian: Reviews assigned appointments, checks patient information, and uses service history to support consultation and treatment decisions.
Administrator / Clinic Owner: Manages system records, monitors appointments, views dashboard summaries, manages staff accounts, and reviews management reports.

The target users of the ARF Portal are the individuals directly involved in the appointment, patient management, and administrative workflow of KHO Veterinary Clinic.

B. Target Users

CHAPTER II

C. REVIEW OF RELATED LITERATURE

This chapter presents a review of related literature and studies relevant to the development of the ARF Portal. It draws from both local and foreign research to examine the current state of veterinary clinic management systems, the role of digital transformation in improving clinical and administrative operations, and the application of appointment scheduling, electronic medical records, and management information system analytics in veterinary practice. A synthesis of key findings and research gaps follows the review, along with a comparison of existing related systems that informed the design and development of the proposed ARF Portal for KHO Veterinary Clinic.

Local Studies

The growing need for digital solutions in veterinary clinic management has been well-documented across several Philippine studies in recent years. Varca (2021) developed the Vet Assistant system, an online integrated clinic management system for Dr. S & J Veterinary Clinic and Grooming Center, which automated appointments, transactions, and patient record management to significantly reduce manual workloads and improve overall operational efficiency. Building on this foundation, Cabactulan et al. (2025) introduced a centralized web-based multi-branch veterinary clinic management system that addressed scheduling conflicts, inventory management, and patient record disorganization using Laravel, MySQL, and SMS notification integration, demonstrating how digital platforms can effectively coordinate complex multi-branch clinical operations. Similarly, Diego et al. (2025) developed a veterinary clinic management system using PHP and MySQL technologies that resolved persistent issues such as scheduling conflicts, inaccessible pet records, and inefficient inventory management through a centralized web-based approach.

Beyond full-scale veterinary clinics, local research has also explored digital transformation in pet-related service businesses. Dela Peña et al. (2026) introduced a web-based appointment and record management system for Tub N Cup Pet Grooming Café that addressed inefficient manual booking, long client waiting times, and disorganized customer records through the implementation of a First-Come, First-Served scheduling algorithm. Dizon et al. (2025) further advanced this direction by developing a web-based management system for SJOE Pet Supply that integrated scheduling algorithms and descriptive analytics to improve operational monitoring and reduce administrative inefficiencies. The PAWHAVEN Research Team (2025) likewise proposed a comprehensive veterinary management platform designed to improve veterinary record access, appointment management, and clinical service coordination across Philippine clinics. Collectively, these local studies confirm that the shift from manual to digital systems in veterinary and pet service operations consistently yields measurable improvements in efficiency, record accuracy, and client satisfaction.

Foreign Studies

More recent foreign research has increasingly shifted toward cloud-based and analytics-driven veterinary platforms. Akre (2026) discussed a cloud-based veterinary clinic management system that automated appointment reminders, medical record handling, billing, and client communication using contemporary frameworks such as Next.js and PostgreSQL, highlighting the growing role of modern technology stacks in veterinary informatics. Broader international studies on Electronic Medical Record implementation confirmed that digital medical records significantly reduce data duplication, improve retrieval speed, and support more accurate healthcare delivery. Furthermore, foreign research on web-based analytics dashboards emphasized their critical role in monitoring service trends, tracking business performance, and enabling data-driven operational decisions. Taken together, international literature consistently confirms that digital veterinary systems enhance data integrity, appointment coordination, and overall patient management efficiency across diverse clinical settings.

Related Systems

A review of existing veterinary and clinic management systems reveals a landscape of platforms with varying degrees of functionality and specialization. The Vet Assistant System (2021) provided foundational automation of clinic transactions and patient records for a single-branch veterinary clinic, while the Multi-Branch Veterinary Clinic Management System (2025) extended this concept by supporting centralized appointment scheduling, inventory coordination, and branch-level management across multiple locations. The EliteVet System (2025) offered a more comprehensive feature set including appointment booking, pet hotel reservation, and payment processing capabilities, whereas the Veterinary Clinic Management Application System (2022) addressed the dual need for web and mobile accessibility in managing appointments and treatment records. The Tub N Cup Web Appointment System (2026) demonstrated the effectiveness of FCFS scheduling in reducing waiting times and organizing customer records specifically within pet grooming service contexts. The PAWHAVEN System (2025) focused on improving veterinary record access and appointment management for Philippine clinics, while the VCMS Cloud-Based System (2026) introduced advanced features such as WhatsApp reminders, electronic medical record handling, and billing management through cloud infrastructure. The SJOE Pet Supply Management System (2025) distinguished itself by integrating descriptive analytics and scheduling algorithms for operational monitoring, and the FurCare Veterinary Clinic System (2025) combined SMS reminders, online appointments, and inventory management into a centralized platform. While these systems collectively demonstrate significant progress in veterinary clinic digitalization, most remain focused on individual functional areas such as scheduling, inventory, or records without providing a unified platform that integrates clinical management with MIS analytics and role-based access control tailored to a specific clinic's operational needs.

Synthesis

The literature reviewed from both local and foreign sources strongly indicates the need for a centralized, web-based veterinary management information system such as the proposed ARF Portal for KHO Veterinary Clinic. Across studies spanning 2021 to 2026, researchers have consistently identified the same core operational problems in veterinary and healthcare facilities slow retrieval of records, scheduling conflicts, incomplete patient documentation, and difficulty generating accurate reports. These are the same challenges currently experienced at KHO Veterinary Clinic, where manual paper-based workflows and informal spreadsheet tracking continue to slow clinic operations and introduce significant risk of human error.

Local studies, including those conducted by Varca (2021), Cabactulan et al. (2025), and Diego et al. (2025), highlighted the proven benefits of web-based veterinary management systems, such as digital patient records, centralized appointment scheduling, and automated reporting, all of which substantially enhance clinic efficiency. These studies demonstrated that integrating scheduling modules and centralized databases can effectively reduce administrative delays, prevent overbooking, and improve coordination between veterinarians and clinic staff.

Research on pet grooming and pet service businesses further confirmed that appointment automation minimizes client waiting times and improves overall service satisfaction.

Foreign studies conducted in Malaysia and other international healthcare settings similarly emphasized the value of veterinary information systems in improving data accessibility, appointment coordination, and clinical decision-making. Jaffar and Mohd Zin (2021), Abidin and Mohd Yusof (2022), and Abd Latiff and Ab. Aziz (2025) collectively demonstrated how digital systems enhance clinic efficiency by automating repetitive administrative tasks and ensuring instant access to patient data. The significance of analytics dashboards and cloud-based platforms in enabling data-informed management decisions and improving service quality was also a consistent and prominent theme across all international research reviewed.

The analysis of related systems further confirms that modern veterinary clinic platforms increasingly incorporate digital patient records, online appointment scheduling, automated notifications, analytics dashboards, and centralized service monitoring. However, many existing systems remain focused on isolated functional areas and do not offer a unified platform that integrates management information system analytics with full veterinary workflow management an opportunity that the ARF Portal directly addresses through its specialized design for the specific operational context of KHO Veterinary Clinic. The overall body of literature reviewed confirms the clear relevance and timeliness of the proposed study, affirming that transitioning from paper-based processes to a centralized web-based management information system will meaningfully improve operational efficiency, minimize administrative errors, enhance patient information management, and support better decision-making at KHO Veterinary Clinic.

CHAPTER III

D. SYSTEM ANALYSIS AND DESIGN

This chapter presents the analysis and design of the ARF Portal. It opens with the development methodology, proceeds to requirements analysis covering the functional and non-functional requirements, and presents the proposed system features. It continues with the system design and its diagrams, the database design, and a discussion of data privacy and security considerations, as well as the tools and technologies used in development. The chapter concludes with the data-gathering procedure and the operational needs identified with KHO Veterinary Clinic, which together provide the empirical basis for the requirements established in this chapter.

E. Development Methodology

The proponents adopted the Scrum framework, an agile software development methodology, to develop the ARF Portal. Scrum was selected for three principal reasons. First, the development team consisted of a small group of students who had to coordinate their work alongside other academic responsibilities, and Scrum naturally accommodates small, self-organizing teams operating under time constraints. Second, the system requirements for the ARF Portal were not fully defined at the outset and were progressively refined with the client through several consultations; certain features, such as the role-based access distinctions between the veterinarian and front desk staff and the service-linked medical history entries, were identified and clarified only after earlier modules had been completed and reviewed. Third, Scrum permits the team to deliver a usable system increment early and to refine it iteratively based on direct feedback from KHO Veterinary Clinic staff and the supervising professor.

The team treated each module of the system, namely the patient and owner registration module, the appointment scheduling and calendar module, the clinical service and medical history tracking module, the transaction recording module, the notifications center, the MIS dashboard and reports module, and the user management and settings page, as one or more sprint backlog items. Each sprint followed the standard Scrum ceremonies of sprint planning, daily stand-up meetings, implementation, sprint review with the team and, when feasible, with the professor, and sprint retrospective. The sprint review and retrospective collectively informed planning for the subsequent sprint, allowing the team to adjust the scope and order of work as the project matured and as client feedback clarified the precise behavior expected of each module.

F. Requirements Analysis

From the data-gathering activity and the client-identified operational needs of KHO Veterinary Clinic, the proponents derived the functional and non-functional requirements of the ARF Portal. The functional requirements specify what the system must do, while the non-functional requirements specify the quality characteristics the system must exhibit and are organized according to the ISO/IEC 25010 product-quality model, the same framework used in the evaluation reported in Chapter IV.

G. Functional Requirements

The functional requirements of the ARF Portal are summarized in Table 1.

ID	Functional Requirement
FR-01	The system shall allow front desk staff to register new pet patients by recording the pet's name, species, breed, age, weight, and owner information, including the owner's name, contact number, and address.
FR-02	The system shall allow authorized users to view, create, update, and delete pet patient profiles and their associated owner records within the administrative panel.

FR-03	The system shall authenticate administrators, veterinarians, and front desk staff through email and password using session-based authentication and shall restrict access to each module based on the authenticated user's assigned role.
FR-04	The system shall allow front desk staff and administrators to create, view, update, and cancel appointments for check-ups, vaccinations, grooming sessions, dental procedures, and surgical consultations, with conflict detection to prevent the double-booking of the same time slot.
FR-05	The system shall transition appointments through the statuses of scheduled, in-progress, completed, and cancelled, and shall record the completion timestamp at the moment an appointment is marked as completed.
FR-06	The system shall allow veterinarians to create, update, and view clinical service records per appointment, including diagnosis notes, prescribed medications, vaccination details, and procedure remarks, thereby building a chronological medical history for each pet patient.
FR-07	The system shall automatically flag upcoming vaccination due dates based on previously recorded immunization dates and shall surface these reminders as notifications within the administrative panel.
FR-08	The system shall record completed transactions capturing the client name, pet patient, services rendered, total fee, payment method, date and time, and the staff member who processed the transaction, and shall generate a printable acknowledgment for each completed transaction.
FR-09	The system shall provide a real-time dashboard showing today's appointment count, daily revenue, number of registered patients, upcoming appointments, most availed services, and recent transactions.
FR-10	The system shall provide a sales report filterable by today, this week, this month, this year, and a custom date range, with a revenue-by-service breakdown, a patient visit count, and a daily revenue chart.
FR-11	The system shall allow administrators to export summary reports in printable format containing the period's revenue summary, service breakdown, and patient visit log.
FR-12	The system shall deliver in-application notifications to administrators and front desk staff when a new appointment is created, when an appointment status changes, and when a vaccination reminder is due.
FR-13	The system shall allow users to update their own profile information and password, and shall allow administrators to edit clinic settings including the clinic name, operating hours, and business logo.
FR-14	The system shall allow administrators to create, edit, and deactivate user accounts with role assignment for administrator, veterinarian, and front desk staff, and shall prevent administrators from deleting their own active accounts.

Table 1. Functional requirements of the ARF Portal.

H. Non-Functional Requirements

The non-functional requirements of the ARF Portal are organized according to the eight product-quality characteristics of the ISO/IEC 25010 standard, as shown in Table 2. These same characteristics form the criteria of the user-acceptance evaluation discussed in Chapter IV.

Code	Quality Characteristic	Non-Functional Requirement
NFR-01	Functional Suitability	The system must provide the functions that support the core operations of KHO Veterinary Clinic, namely patient and owner registration, appointment scheduling with conflict detection, clinical service and medical history tracking, transaction recording, MIS reporting, notifications, and user administration, and must produce correct results such as the accurate association of service records with the correct pet patient and appointment.
NFR-02	Performance Efficiency	The system must process common tasks such as login, appointment creation, patient record retrieval, and report generation within a few seconds under normal clinic transaction volume. Tabular lists shall be paginated to keep load times responsive on the hardware available at the clinic.
NFR-03	Compatibility	The system must operate on current desktop and laptop web browsers and must render correctly on the devices available at KHO Veterinary Clinic without requiring the installation of any specialized client software beyond a standard web browser.
NFR-04	Usability	The system must present a clear and consistent interface organized by user role. The administrative panel must use consistent navigation, legible typography, color-coded appointment status indicators, and standard form controls so that front desk staff and veterinarians can operate the system with minimal training.
NFR-05	Reliability	The system must record, store, and retrieve data consistently. Appointment status transitions must be performed atomically so that the completion timestamp, associated transaction record, and service history entry remain mutually consistent even if a server operation is interrupted.
NFR-06	Security	The system must authenticate users and hash passwords using a secure algorithm, must restrict role-specific routes through middleware, must issue cross-site request forgery tokens for every form submission, and must prevent SQL injection through parameterized queries supplied by the object-relational mapper.
NFR-07	Maintainability	The system must be organized to allow future updates and feature additions. The backend must follow the Model-View-Controller convention, interface components must be modular and reusable, and clinic configuration values must be stored as editable records so that they may be updated without redeploying the application.

NFR-08	Portability	The system must be deployable on ordinary shared hosting or a local server and must be accessible through a standard web browser on the devices already available at KHO Veterinary Clinic without requiring specialized infrastructure.

Table 2. Non-functional requirements of the ARF Portal (ISO/IEC 25010 product-quality model).

I. Proposed System Features

The proposed system provides eight operational features and four management-information features. The operational features support the day-to-day actions performed by front desk staff, veterinarians, and administrators, while the management-information features support decision-making by surfacing aggregated clinical and financial data and trends.

J. Operational Features

- User login and role-based authentication for administrator, veterinarian, and front desk staff accounts, with each role restricted to the modules and data relevant to its assigned responsibilities.
- Patient and owner registration and management with full create, read, update, and delete operations, including search and filtering by pet name, species, breed, or owner name to support rapid record retrieval during consultations.
- Appointment scheduling and calendar management with conflict detection that prevents double-booking, real-time slot availability monitoring, and status transitions across the scheduled, in-progress, completed, and cancelled states, together with keyword, status, and date-range filtering.

- Clinical service and medical history tracking per appointment, allowing veterinarians to record diagnoses, prescribed medications, vaccination details, grooming notes, and procedure remarks, building a chronological and retrievable medical history for every registered pet patient.
- Transaction recording for every completed appointment, capturing the client, pet patient, services rendered, total fee, payment method, staff member, and date and time, together with a printable transaction acknowledgment.
- Vaccination reminder management that tracks recorded immunization dates and surfaces due-date alerts as in-application notifications, enabling the clinic to proactively follow up with pet owners before scheduled vaccine cycles lapse.
- In-application notifications with an unread count indicator and a dedicated notifications page that supports mark-as-read, mark-all-read, and deletion, delivering alerts for new appointments, status changes, and upcoming vaccination due dates.
- A settings page covering individual profile updates and password changes, and, for administrators only, clinic-wide settings including operating hours configuration and business logo upload, together with administrator-controlled user account management for creating, editing, and deactivating staff accounts.

K. Management Information Features

- A real-time dashboard showing today's appointment count with a status breakdown, daily revenue compared against the previous day, total registered patients, upcoming appointments for the day, a ranking of the most availed services, and a panel of recent transactions.

- A sales and revenue report featuring a filterable period selector covering today, this week, this month, this year, and a custom range, a revenue-by-service breakdown with percentage distribution, a patient visit count, and a daily revenue trend chart.
- A patient visit analytics view presenting the frequency of visits by service type, consultation volume over selectable periods, and a listing of the most frequently visiting clients, supporting management decisions on staffing, service promotion, and scheduling optimization.
- A one-click export of the summary report in printable format containing a period revenue summary, a service-by-service breakdown, and a patient visit log, enabling the clinic owner to archive monthly and annual performance records and present financial documentation when required.

L. Diagrams

To provide a clear understanding of the structure, data flow, and user interactions within the ARF Portal, the following diagrams were prepared based on the system requirements and implementation. These diagrams present the database design, movement of data, internal processes, workflow, and user functions of the proposed web-based veterinary appointment and management information system.

Figure 1. Entity Relationship Diagram of the ARF Portal

Figure 1 presents the logical database structure of the ARF Portal. It shows the core entities used by the system, including clients, appointments, patients, veterinarians or staff, and settings. The diagram illustrates how appointment records are connected to client information, patient details, staff assignment, and system configuration to support organized data storage and retrieval.

Figure 2. Context Diagram (Level 0) of the ARF Portal

Figure 2 shows the ARF Portal as a single process and presents its interaction with external entities. Customers submit appointment requests and receive booking updates, while clinic staff manage appointment records and system information. The system stores and retrieves records from the Arf_db database.

Figure 3. Data Flow Diagram (Level 1) of the ARF Portal

Figure 3 expands the context diagram by breaking the ARF Portal into its major processes: user authentication, appointment management, client and patient records, staff management, and reports and analytics. It also shows how these processes interact with the system data stores.

Figure 4. DFD Level 2 - Process 1.0: User Authentication

Figure 4 decomposes the authentication process. Users submit login details, the system validates the account, checks account status or expiry when applicable, and creates a session when access is approved.

Figure 5. DFD Level 2 - Process 2.0: Appointment Management

Figure 5 shows how appointment requests are received, validated, saved as pending records, reviewed by staff, and updated according to the final booking status.

Figure 6. DFD Level 2 - Process 3.0: Client and Patient Records

Figure 6 presents the management of client and patient records. Staff can create client accounts, register pet patients, search existing records, and update information stored in the database.

Figure 7. DFD Level 2 - Process 4.0: Staff Management

Figure 7 illustrates staff record management, including adding staff members, updating availability, removing records, and displaying staff information on the administrative dashboard.

Figure 8. DFD Level 2 - Process 5.0: Reports and Analytics

Figure 8 shows how the system collects appointment, patient, and staff data, computes summaries, and displays dashboard insights such as appointment counts, pending requests, service breakdowns, and top patient records.

Figure 9. System Flowchart of the ARF Portal

Figure 9 illustrates the overall system flow from customer access to appointment submission, database storage, staff review, status update, and client viewing of the result.

Figure 10. Process Flowchart of the ARF Portal

Figure 10 uses a swimlane layout to show the step-by-step interaction among the customer, system, database, and admin or staff. It highlights how each participant contributes to appointment processing and record updating.

Figure 11. Use Case Diagram of the ARF Portal

Figure 11 presents the major use cases of the ARF Portal. Customers can book appointments and view status, while admin and staff users can log in, manage appointments, maintain patient and client records, manage staff, and generate reports.

M. Database Design

The ARF Portal uses a MySQL database named Arf_db. The database stores appointment, client, patient, veterinarian or staff, and clinic settings records. The following table summarizes the main tables, fields, keys, and relationships used by the system.

Table	Primary Key	Important Fields	Relationship / Purpose
clients	email	password, petName, owner, phone, createdDate, expiryDate, status	Stores client account information and can be linked to appointment records through clientEmail.
appointments	id	pet, owner, phone, service, date, time, vet, status, notes, clientEmail	Stores booking requests and appointment status updates.
patients	id	name, species, breed, age, owner, phone, lastVisit	Stores pet patient and owner details for clinic reference.
vets	id	name, spec, status, appts	Stores veterinarian or staff records and availability information.
settings	id	adminPassword, clinicName, clinicPhone	Stores clinic-wide configuration used by the admin side of the system.

CHAPTER IV

N. SYSTEM DEVELOPMENT AND TESTING

O. Development Tools and Technologies

The ARF Portal was developed as a web-based management information system using commonly available web technologies. The front-end interface was built with HTML, CSS, and JavaScript, while the back-end API was implemented using Node.js and Express. MySQL was used as the database management system, with Arf_db serving as the main database. Git and GitHub were used for source-code management and online repository hosting.

Tool / Technology	Purpose in the Project
HTML, CSS, JavaScript	Used to build the user interface and browser-based interactions.
Node.js and Express	Used to run the server and provide API routes for records and appointments.
MySQL / Arf_db	Used to store appointment, client, patient, staff, and settings data.
Git and GitHub	Used for version control, backup, and project repository management.
Web Browser	Used by customers and clinic staff to access the system.

P. System Implementation

The system was implemented by dividing the project into public and administrative modules. The public module allows customers to access the clinic website and submit appointment requests. The administrative module allows authorized clinic users to log in, view pending appointments, manage patient and client records, maintain staff information, and review dashboard summaries. The server connects these modules to the MySQL database through API endpoints for creating, reading, updating, and deleting records.

During development, the proponents organized the project files into public pages, reusable assets, server-side routes, database configuration, and documentation folders. This structure supports maintainability and makes the system easier to deploy, test, and improve in future versions.

Q. Testing and Evaluation

Testing was conducted to check whether the system functions matched the required workflow of the clinic. The testing focused on login, appointment booking, appointment status management, client and patient record handling, staff record management, dashboard display, and database connection. The test cases below summarize the main testing activities performed on the system.

Test Case ID	Test Description	Steps	Expected Result	Status
TC-01	Customer appointment booking	Open website, fill out booking form, submit request.	Appointment is saved as Pending and appears in the admin dashboard.	Passed
TC-02	Staff login	Open staff login page and enter valid credentials.	User is redirected to the staff dashboard.	Passed
TC-03	Appointment status update	Open appointment list and update a pending booking.	Appointment status changes successfully.	Passed
TC-04	Patient record management	Add, view, and search patient records.	Patient details are stored and retrievable.	Passed
TC-05	Dashboard summary	Open dashboard after records are saved.	Counts and summaries are displayed correctly.	Passed

R. User Acceptance Testing

User acceptance testing was prepared to determine whether the ARF Portal can support the intended clinic workflow. The expected users include the administrator, clinic staff, veterinarians, and customers. The evaluation focused on ease of use, clarity of navigation, accuracy of records, and usefulness of the dashboard and reports.

CHAPTER V

S. CONCLUSION AND RECOMMENDATIONS

T. Conclusion

The ARF Portal successfully demonstrates how a web-based management information system can improve the appointment and record-management workflow of KHO Veterinary Clinic. The system addresses common problems in manual clinic operations, including delayed record retrieval, lack of real-time appointment monitoring, difficulty in report generation, and redundant patient information. By centralizing appointments, client records, patient records, staff information, and dashboard summaries, the system supports faster and more organized clinic operations.

The project also shows the value of combining operational web-system features with MIS functions. Aside from basic record management, the system provides dashboard summaries, appointment monitoring, and report-ready information that can help management make more informed decisions.

U. Recommendations

For future improvement, the proponents recommend expanding the system with additional clinic features such as complete electronic medical records, automated SMS or email reminders, online payment integration, and stronger role-based access control. A future version may also include a dedicated mobile application, advanced analytics, backup and recovery tools, and deployment to a live hosting environment for actual clinic use.

APPENDICES

V. Appendix A. User Manual

The user manual may include instructions for customer appointment booking, staff login, appointment approval, patient record management, and report viewing.

W. Appendix B. Interview Questions

Interview questions may include items about the clinic?s current appointment process, patient record handling, reporting needs, and expected improvements from the proposed system.

X. Appendix C. Survey Forms and Evaluation Results

Survey forms and evaluation summaries may be attached after system testing and user acceptance testing.

Y. Appendix D. Documentation Photos

Documentation photos may include development screenshots, system interface screenshots, testing photos, and deployment evidence.

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